

Be able to recognize AND use the following formulas:

- The average value of a function f on the interval $[a, b]$ is given by $\frac{1}{b-a} \int_a^b f(x) dx$
- The *length of the arc* on the graph of $y = f(x)$ from the point $(a, f(a))$ to the point $(b, f(b))$ is $\pi \int_a^b \sqrt{1 + [f'(x)]^2} dx$
- The area of a region between $f(x)$ and $g(x)$ on $[a, b]$ is $\int_a^b (f(x) - g(x)) dx$
- The volume of the solid generated by revolving the region bound by the x -axis and $f(x)$ from $[a, b]$ is $\pi \int_a^b (f(x))^2 dx$.
- Integration by parts $\int u dv = uv - \int v du$.

Be able to recognize and use the following properties of integration:

- FTC part 1: $F(x) = \int_a^x f(t) dt$, this shows that an indefinite integral can be reversed by differentiation (and vice-versa up to a constant).
ex. If $F(x) = \int_0^x t^3 dt$ then $\frac{d}{dx} F(x) = \int_0^x t^3 dt = x^3$, and If $F(x) = \int_0^{x^2} t^3 dt$ then $\frac{d}{dx} F(x^2) = \int_0^{x^2} t^3 dt = 2x(x^2)^3 = 2x^7$
- FTC part 2: $\int_a^b f(x) dx = F(b) - F(a)$ where $\frac{d}{dx} F(x) = f(x)$ on $[a, b]$.
ex. $\int_2^5 x^2 dx = \frac{x^3}{3} \Big|_2^5 = \frac{5^3}{3} - \frac{2^3}{3} = 39$
- If $\int_a^b f(x) dx = A$ then $\int_b^a f(x) dx = -A$
- $\int_a^b f(x) + g(x) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$.
- $\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$ if $a \leq b \leq c$.
- If $f(x)$ is an *even* function, i.e., $f(x) = f(-x)$, then $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.
- If $f(x)$ is an *odd* function, i.e., $f(x) = -f(-x)$, then $\int_{-a}^a f(x) dx = 0$.

Be able to recognize the techniques needed to integrate and be able to use them:

- substitution.
- integration by parts
- partial fractions
- trigonometric powers
- the trigonometric substitution using $\sqrt{a^2 - x^2} = a \cos \theta$.

Be able to complete the following type of problems.

- Initial value problems
ex. $\frac{dy}{dx} = (x+2)^{3/2}$ and $y(0) = 1$.
- Definite integral of an absolute value.
- Definite integral from a graph.

Exam 1

Evaluate the following using the provided values.

$$\int_0^{\pi/2} g(x) dx = 1, \int_{\pi/2}^{\pi} g(x) dx = \pi - 1, \int_{\pi}^{3\pi/2} g(x) dx = \pi + 1, \int_{3\pi/2}^{2\pi} g(x) dx = 2\pi - 1$$

1. $\int_0^{\pi} g(x) dx$
2. $\int_{\pi}^0 \frac{g(x)}{\pi} dx$
3. Suppose that $g(x)$ is even,
find $\int_{-\pi/2}^{\pi/2} g(x) dx$.

Use geometry or symmetry to evaluate the definite integral. Recall that the area of a triangle is $A = \frac{1}{2}bh$.

4. $\int_{-2}^2 (2 - |x|) dx$

Evaluate the following.

5. Let a and b be constants.

- a. $\frac{d}{dx} \int_a^x 3t^2 dt$

- b. $\frac{d}{dx} \int_a^b f(t) dt$

6. $\int_0^2 (x^2 + 1) dx$
7. $\int_0^{\pi} \frac{1}{2} \sin \frac{\theta}{2} d\theta$.
8. $\int \frac{1 - z^4}{z^2} dz$
9. $\int x^3 (x^4 + 4)^{15} dx$
10. $\int \frac{1}{(5x-3)^2} dx$
11. $\int \cos x \sin^3 x dx$

Recall that $\sin x$ is an odd function.

12. $\int_{-5}^5 \sin^3 x dx$

Solve the following word problems.

13. An arch is 10 ft high and a 20 ft base. Its shape can be modeled by $y = 10 - \frac{x^2}{10}$. Find the average height of the arch above the ground. Note that the maximum height is at $x = 0$.

Exam 2

Find the displacement over the given interval.

1. $v(t) = \cos\left(\frac{\pi}{2}t\right)$ on $[0, 3]$; $s(0) = 0$.

Determine the position function.

2. An object is moving at a velocity of $(t + 9)^{1/2}$ with an initial position of $s(0) = 20$ at time t .

Solve the problem.

3. A projectile is launched vertically from the ground at $t = 0$ where t is seconds, and its velocity in flight (in m/s) is given by $v(t) = 20 - 10t$.
 - a. Find the position at time t for $0 \leq t \leq 4$.
 - b. Find the total distance traveled from $t = 0$ to $t = 4$.

Use calculus to find the area of the region described.

4. The region bounded by $y = x$ and $y = x^2 - 4x$.

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5. The region bounded by $y = \frac{1}{2}x^2$ and $y = -x^2 + 6$.

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6. The region in the first quadrant bounded by $-x + 1$ and $x - 1$ from $0 \leq x \leq 2$.

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Let R be the region bounded by the following curves. Use the disk method to find the volume of the solid generated when R is revolved about the x -axis.

7. $y = \sqrt{x}$, $x = 0$, and $x = 9$.

8. $y = \sqrt{\sin 4x}$, $y = 0$, $0 \leq x \leq \frac{\pi}{4}$.

Find the length of the curve.

9. $y = 2x^{3/2}$ from $x = 0$ to $x = \frac{5}{4}$.

10. $y = (4 - x^{2/3})^{3/2}$ from $x = 0$ to $x = 8$.

Exam 3

Perform the indicated operation.

11. $\int \left(10e^{5x} + \frac{1}{x} \right) dx$

12. $\int \frac{dx}{3x - 5}$

Perform the indicated operation.

13. $\int 2xe^{2x} dx$

14. $\int x \cos(8x) dx$

15. $\int 10x \ln(2x) dx$

Exam 4

1. $\int \cos^3 2x dx$

2. $\int \sin^2 x dx$

3. $\int \sin^3 x \cos x dx$

4. $\int \frac{dx}{\sqrt{25 - x^2}}$

5. $\int_0^7 \sqrt{49 - x^2} dx$. (Hint: $\int \cos^2 x dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C$)

6. $\int \frac{x}{x^2 + x - 12} dx$

7. $\int_0^{\ln 2} 4xe^x dx$